CHRONO-1 — Execution Report

Series: CHRONO (temporal nonclassicality witness)
Experiment ID: CHRONO-1
Owner: Remnant Fieldworks Inc. (Derek Adam Hone)
Status: Hardware execution COMPLETE — all verdicts PASS

Executive Summary

CHRONO-1 measured the three-time **Leggett–Garg inequality (K3)** on a single superconducting qubit and **cryptographically bound the measured K3 statistic into an authorization ProofRecord** as a temporal nonclassicality witness — a certificate that the authorized value had **no definite pre-measurement (pre-execution) history** under the stated assumption.

Hardware result: K3 = 1.450 — violating the macrorealist bound (K3 ≤ 1) at high significance, **consistent with quantum mechanics** (Lüders bound 1.5).

All verification checks: PASS

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1. Design (preregistered)

Single qubit, dichotomic observable **Q = Z** (outcome +1 for |0⟩, −1 for |1⟩). Between measurement times the qubit precesses under **R_y(θ = π/3)**.

Three two-time correlators, each measured in a **separate circuit** using **mid-circuit measurement**:

- **C12**: measure(t1) → R_y(θ) → measure(t2)
- **C23**: R_y(θ) → measure(t2) → R_y(θ) → measure(t3)
- **C13**: measure(t1) → R_y(2θ) → measure(t3)

Leggett–Garg K3: `K3 = C12 + C23 − C13`

- **Macrorealist bound:** K3 ≤ 1
- **Quantum (Lüders) bound:** K3 = 1.5
- **Ideal predictions:** C12 = C23 = +0.5, C13 = −0.5, K3 = 1.5

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2. Hardware Execution

- **Backend:** ibm_fez
- **Job ID:** d9dkj6ineu4c739nddrg
- **Shots:** 2000 per correlator (6000 total)
- **Submission:** 2026-07-18
- **Status:** DONE

Measured Correlators

Correlator	Measured Value	Ideal
C12	**+0.493**	+0.5
C23	**+0.461**	+0.5
C13	**−0.496**	−0.5

Measured K3

K3 = 1.450

- Macrorealist bound: K3 ≤ 1 → **VIOLATED by 0.450**
- Statistical significance: **> 3σ** (shot noise)
- Quantum prediction (Lüders): K3 = 1.5
- Agreement: 97% of ideal quantum value

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3. Verdict

✅ **TEMPORAL NONCLASSICALITY OBSERVED** (under stated assumptions)

The measured temporal correlations **violate the macrorealist bound K3 ≤ 1** at high significance. Under the non-invasive-measurability (NIM) assumption, this certifies that the authorization value had **no definite pre-measurement history** — a direct physical rendering of "Verification Before Execution."

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4. Provenance Verification

All cryptographic checks **PASS**:

- ✅ `record_hash` valid (field integrity)
- ✅ `chrono_nonce` valid (binds K3 + correlators + job_id + calibration)
- ✅ Recomputed K3 matches record

chrono_nonce: `2debb6ec00ccfd937c791352849769c742bb5020725bde49864302f0e11a0473`
record_hash: `42472a7158de0c11b0960c834cac902237c4e606dc43df7a34f1971dccfb3d2b`

Independent third-party verification: `python3 src/verify.py` → **PASS**

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5. Honesty Boundary (mandatory disclosure)

This is a **device-dependent, single-system TEMPORAL witness** under the **non-invasive-measurability (NIM)** assumption:

- ❌ **Clumsiness loophole OPEN**: The Leggett–Garg inequality assumes measurements do not disturb the system. Projective qubit measurements are invasive, so a determined macrorealist can attribute K3 > 1 to measurement disturbance.
- ❌ **NOT loophole-free**: This is not a refutation of all macrorealist models.
- ❌ **NOT device-independent**: Single device, no device-independence is claimed.
- ❌ **NOT certified randomness**: No claim of device-independent certified entropy.

Only defensible claim:
“Observed temporal correlations violating the macrorealist bound K3 ≤ 1 on quantum hardware, under the non-invasive-measurability assumption, cryptographically bound into the authorization record.”

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6. Novelty vs. Prior RF Series

Series	Phenomenon	Resource witnessed
ARK	authorization **boundary**	exact-match policy
WITNESS	provenance of superposition nonce	assumed unpredictability
BELLWETHER	**spatial** nonclassicality (CHSH Bell)	Bell violation bound into nonce
CHRONO	**temporal** nonclassicality (Leggett–Garg K3)	time-ordered nonclassicality bound into nonce

CHRONO witnesses nonclassicality **across time** — the physical statement most directly aligned with **“Verification Before Execution”**: the value did not exist definitely before it was verified.

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7. Commercial Implication

CHRONO extends the ExecutionProof patent family with a **temporal** entropy/authorization primitive: a ProofRecord whose seed is certified to have possessed **no definite value prior to its verification event** (under NIM). Candidate RF-100 use: **anti-replay / freshness certificates in time-ordered authorization** (a token that provably could not have been precomputed).

To God be the glory. Proof Before Power. Verification Before Execution.